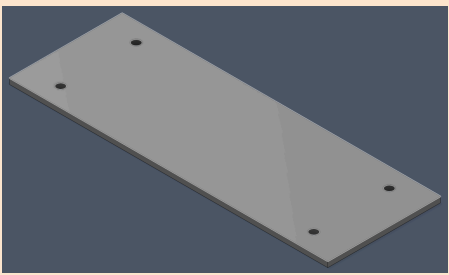
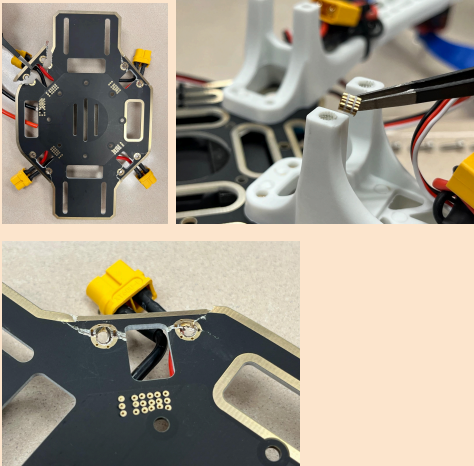
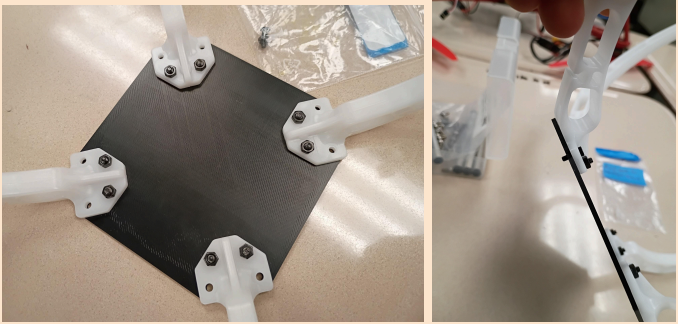
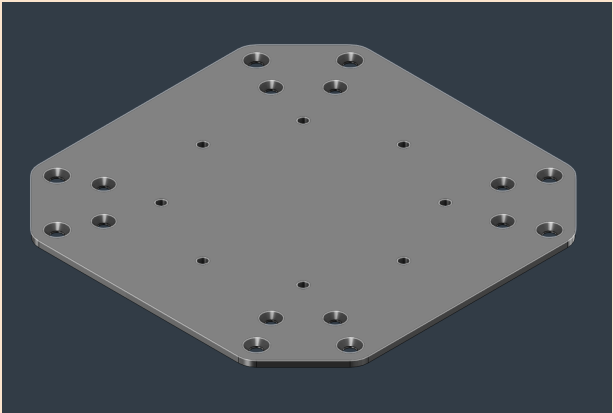


Belleville Engineering Notebook

1-3-2026	We created our first git repository. We started coding by implementing the mpu6050 accelerometer + gyroscope. One of our team members spent a while implementing the Runge-Kutta Order 4 algorithm.
1-19-2026	Created a new repository initialized from Raspberry Pi and brought all code from the original repository over.
1-22-2026	Finished the initial implementation of the Runge-Kutta numerical integration algorithm. Testing on the Raspberry Pi proved that our orientation tracking is working well. Position tracking still needs some work, though.
1-28-2026	We started building the drone. We started by laying out all of the parts and making sure everything was here. We started watching the tutorials to familiarize ourselves with the build.
2-30-2026	We started assembling the drone. We attached the arms to the upper and lower chassis PCBs. We also mounted all 4 of the brushless DC motors.
2-4-2026	We stuck the stabilizer onto the upper chassis and wired together the main components after adding the flight controller.
2-6-2026	We used Mission Planner to calibrate the flight controller and radio on the drone.
2-11-2026	We watched the tutorial to learn how to safely charge our team's LiPo battery. After a while of charging, we powered up our drone and gave it a quick 5-second test flight to verify that we had built it correctly.

2-13-2026  raspberrypi platform v1	We laid out all of the sensors and components that go with the Raspberry Pi and designed a platform for them all to sit on.
2-18-2026	The camera on the Pi was tested, and we were able to capture a picture after every click and display it on the website. We also sped up the amount of time each photo took to show, raising the fps.
2-20-2026	We programmed the sample crane motor. We then tested our code on the Raspberry Pi after wiring the H-bridge motor driver to the GPIO pins, and verified that our implementation worked.
2-25-2026  Drone damage pictures	We tested and broke the drone. The arms and power distribution were rendered unusable. This occurred because we could only screw on the legs with 2 holes.

2-27-2026	We began thinking of and working out solutions to our problem with the legs. Brainstorming began. We were looking for ways to prevent the legs from bending so much as to break other parts of the drone.
3-04-2026	The new drone kit came in, and we began to fix the drone. We also started designing a 3-D printed part to be able to better attach the legs.
3-06-2026 	We continued to put the drone together. We mounted the legs to the 3-D printed platform. We found out that we needed longer bolts to attach the platform to the arms and the rest of the drone.
3-11-2026	We finished assembling the main part of the drone (except the feet). We reconnected every cable and zip tied everything down.
3-13-2026 	We figured out that instead of having to find longer bolts for our drone leg platform, we can just use the 8 extra random holes on the bottom of the drone. After chamfering the other 2 holes per leg to allow 4 M3 bolts to hold each leg down and adding the 8 mounting holes we are ready to print. We also test-flew the drone.
3-18-2026	Added code that calculates altitude from pressure.

3-20-2026	Test flight of the drone, worked on precision landing (on a target on the floor). And practiced control of the drone.
3-25-2026	Added motor for sample retrieval, placed camera under drone's platform for the best viewing angle.
3-26-2026	Customized the HTML page. Tested the sensors with our code. Noticed that the motor and moved e3
3-27-2026	Moved motor for the sample retrieval to directly under the drone, also moved the camera to adjust to that change.